

Ray Optics and Optical Instruments

Light, Ray Optics and the Plane Mirror

NCERT sections covered: 9.1, 9.2

Nature of light (NCERT 9.1)

Light lets us see the objects around us — in a dark room nothing is visible until the light is switched on.

Light has a **dual nature**:

Wave nature	Particle nature
explains interference and diffraction	explains the photoelectric effect (Einstein, Nobel Prize)
wave theory — Huygens	particles of light are photons , <i>bundles of energy</i> ; corpuscular idea — Newton

Board slip: the board writes "(Maxwell)" under the particle column. The teacher says **Newton** in class, and that is correct. Newton proposed the corpuscular (particle) theory and Einstein's photoelectric work introduced photons. Maxwell's contribution was the *electromagnetic wave* theory of light, which belongs in the wave column.

Ray optics (NCERT 9.1)

Ray optics is a limiting case of wave optics. Light *can* bend round a corner, but only when the obstacle is extremely small. NCERT gives the reason: the wavelength of light (400–750 nm) is tiny compared with everyday objects. So for ordinary objects light travels in a straight line — **rectilinear propagation of light** — and does not bend while it stays in one medium. A straight-line path is a **ray**, and a bundle of rays is a **beam**.

Reflection, absorption, transmission

Light is a form of energy. When it falls on a surface, three things can happen. It can be **reflected**, **absorbed** by the material, or **transmitted** (it goes through to the other side). How much of each happens depends on the surface:

- **Mirror** — highly reflecting; absorption and transmission are negligible. Mirrors are coated with highly reflective metals such as **silver or aluminium**. The back (non-reflecting) side is drawn with slant hatching.
- **Glass slab** — reflection is negligible and transmission is maximum (this comes back under refraction).
- **Wall painted black** — most of the light is absorbed. A candle in a dark room with black walls gives poor illumination, but placed in front of a mirror it gives more.

Laws of reflection (NCERT 9.2)

Draw the **normal** at the **point of incidence**.

- **Angle of incidence** $\angle i$ — the angle made by the incident ray with the normal.
- **Angle of reflection** $\angle r$ — the angle made by the reflected ray with the normal; the reflected ray stays in the same medium.

1. $\angle i = \angle r$.
2. The incident ray, the normal *at the point of incidence*, and the reflected ray all **lie in the same plane**.

The teacher explains the second law with the three coordinate planes (xy, yz, xz). If the incident ray and the normal lie in one plane, the reflected ray is in that same plane — it can never jump into another plane. NCERT adds that both laws hold at every point of *any* surface, plane or curved.

Plane mirror — characteristics

1. **Size of image = size of object.**
2. The image is formed **on the other side of the mirror**, at the **same distance from the mirror** as the object. An object 20 cm in front gives an image 20 cm behind.
3. The image is **virtual and erect**. *Virtual* means it is formed by rays that only *appear* to come from a point.
4. The image is **laterally inverted**. Take y as vertical and x as horizontal. There is no inversion along y (your head does not go down), but left and right swap along x . The letter **P** or **B** comes out reversed.

Locating an image — ray diagram

To locate a point of the image you need at least two rays. Any number of rays leave each point of the object. Two are enough, and drawing more only clutters the figure.

- Take two rays from **B**. Each one reflects with $\angle i = \angle r$ about its own normal. Produce the reflected rays **backwards** (dashed lines) — they meet at **B'** behind the mirror.
- Do the same from **A** to get **A'**, then join **A'B'**. Using the two end points takes care of every point in between.
- The dashed lines are only *apparent* paths — no light actually comes from behind the mirror, and there is no source there. That is why a plane-mirror image is **virtual**.

Working the same construction point by point for the letter **P** shows lateral inversion directly. Points close to the mirror stay close to it in the image.

Note on this lecture's transcript: gemini-3.5-flash was in a 503 outage and then out of daily quota when this chapter was processed, so the audio was transcribed by gemini-3.5-flash-lite in overlapping five-minute windows, joined where the overlapping speech matches word for word. Lite's clock ran fast (by up to 2x) inside a window; each window was rescaled to its true length, so times quoted here are approximate (roughly ± 30 s). Every gap of more than 30 s was re-transcribed on its own to confirm whether speech was missing. The first five

minutes of this lecture did come from gemini-3.5-flash before its quota ran out. Equations are taken from the board frames, not from the transcript.

Verify these spans

- [17:29-18:12] The transcript gives the set-up of the plane-mirror ray diagram twice in a row ('let us suppose this is the mirror... A and B is the object... first ray like this'). It reads as a transcription repeat rather than new teaching. Nothing is missing on either side of it, and the board frame matches the single construction described in the note.

Spherical Mirrors: Pole, Centre of Curvature, Principal Axis, Focus

NCERT sections covered: 9.2, 9.2.2

Spherical mirrors (NCERT 9.2)

A steel spoon is the everyday model. Its inner, hollow side is **concave** and its outer, bulging side is **convex**. Mirrors that are a part of a sphere are **spherical mirrors**.

- **Concave mirror** — a spherical mirror whose reflecting surface is **curved inwards** (the hollow side reflects; the outside is non-reflecting).
- **Convex mirror** — a spherical mirror whose reflecting surface is **curved outwards** (the bulging side reflects).

In diagrams the slant hatching marks the non-reflecting back.

Terms for a spherical mirror (NCERT 9.2)

1. **Pole (P)** — the centre of the reflecting surface of a spherical mirror.
2. **Centre of curvature (C)** — the centre of the sphere of which the spherical mirror is a part. Cut a hollow ball and silver one piece on the inside: that piece is a concave mirror, and the ball's centre is its C. Unlike P, **C does not lie on the mirror**. For a concave mirror C is on the reflecting side; for a convex mirror it is behind, on the non-reflecting side.
3. **Radius of curvature (R)** — the radius of the sphere of which the spherical mirror is a part. Here $PC = R$.
 - C is a **point**; R is a **length** (SI unit: metre). "Length of the centre of curvature" is a meaningless question.
4. **Principal axis** — the line passing through C and P. It is the normal to the mirror at its pole.

Normal to a spherical surface

For a plane surface the normal is simply the perpendicular. For a spherical surface, **the normal at any point is the line through that point which, produced back, passes through the centre of curvature C** — that is, the radius at that point. The laws of reflection ($\angle i = \angle r$, one plane) still hold at every point, measured from this normal.

So for a ray parallel to the principal axis, draw the normal through C. Then $\angle i = \angle r$ fixes where the reflected ray must go. On a concave mirror it bends **towards** the axis; it cannot go anywhere else.

Principal focus and focal length (NCERT 9.2.2)

Burning paper with a concave mirror: point a concave mirror at the Sun and hold bits of paper in front of it. After a few minutes they catch fire. Rays from a very distant object (the Sun) arrive **parallel**, and the mirror concentrates them at one point.

- **Principal focus of a concave mirror (F)** — when parallel rays are incident on a concave mirror, after reflection they **meet** at a point on the principal axis; this point is the principal focus.
- **Principal focus of a convex mirror (F)** — when parallel rays fall on a convex mirror, after reflection they **appear to meet** (diverge from) a point behind the mirror; this point is the focus. The image idea here is *virtual*.
- **Focal length (f)** — the distance between the pole and the focus, $PF = f$.

The focus is the **midpoint of P and C**, $PF = FC$, so

$$f = \frac{R}{2}$$

(for a mirror of small aperture — proved in a later lecture).

Drawing tip for exams (convex mirror): mark F first, behind the mirror, and then draw each reflected ray as a straight line from F through the point where the ray meets the mirror. If C would fall off the page, move it closer, but keep $PF = FC$.

Worked questions

Q. The radius of curvature of a concave mirror is -20 cm. Find its focal length.

$$f = \frac{R}{2} = \frac{-20}{2} = -10 \text{ cm}$$

(The minus sign comes from the sign convention, taught in a later lecture.)

Q. What are the focal length and radius of curvature of a plane mirror? (*Left for the class to think about.*) A plane mirror is part of a sphere of **infinite** radius, so $R = \infty$ and $f = R/2 = \infty$.

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Verify these spans

- [14:06–14:45] The teacher says 'convex lens' twice while drawing the convex-mirror version of the labelled figure. The board frame shows a convex mirror, and the note follows the board.
- [36:31–36:52] The closing question (focal length and radius of curvature of a plane mirror) is left for the class to think about; the answer in the note (both infinite) is not from the lecture.

Important Rays for Drawing Mirror Ray Diagrams

NCERT sections covered: 9.2, 9.2.2

Why special rays?

From a point **B** of an object, countless rays strike the mirror. For any one of them you *could* draw the normal through C and apply $\angle i = \angle r$. To avoid that work every time, four rays whose reflected path is already known are used. **Any two** of them fix the image: the image of a point is where two reflected rays intersect, or where they *appear* to intersect when produced backwards.

The four important rays (NCERT 9.2.2)

1. A ray parallel to the principal axis passes through the focus after reflection.

- Concave mirror: the reflected ray actually passes through F.
- Convex mirror: the reflected ray **appears to diverge from F** behind the mirror (draw the backward extension dashed).

This is why a concave mirror is **converging** in nature (a parallel beam meets at F) and a convex mirror is **diverging** in nature (a parallel beam spreads as if from F).

2. A ray passing through the focus becomes parallel to the principal axis after reflection.

- Concave mirror: a ray through F reflects parallel.
- Convex mirror: a ray **directed towards F** (behind the mirror) reflects parallel to the axis.

3. A ray passing through the centre of curvature retraces its path after reflection.

Compare a plane mirror hit at normal incidence: $\angle i = 0$ and so $\angle r = 0$, and the ray comes straight back. A ray through C travels along a radius, so it meets the mirror *along the normal* and returns the same way. For a convex mirror, the ray **directed towards C** retraces its path. *Quiz form:* if a figure shows a ray striking a mirror and retracing its path, the point it was aimed at is **C** (not F or P).

4. A ray incident obliquely at the pole P is reflected obliquely with $\angle i = \angle r$. At P the principal axis is itself the normal (it passes through C), so the reflected ray makes the same angle with the axis on the other side. This is true for both concave and convex mirrors.

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Images Formed by a Concave Mirror

NCERT sections covered: 9.2.3

Image formation by a concave mirror

The object is brought in from infinity towards the mirror: infinity $\rightarrow$ beyond C $\rightarrow$ at C $\rightarrow$ between F and C $\rightarrow$ at F $\rightarrow$ between P and F. The image changes in each case.

How to draw: draw the mirror, then the principal axis through its centre, and mark P, F and C, with C at twice the distance of F. Use a scale and put arrows on the rays. Solid lines mean real rays; dashed lines mean rays produced backwards (virtual). *Size tip:* where the image will be smaller (object beyond C), draw the object a bit large. Where the image will be bigger (object between F and C), draw it a bit small. For the object at C, draw the inverted, same-size image first and then fit the rays to it.

Position of object	Position of image	Nature and size
At infinity	At F	Real, inverted, highly diminished (point-sized)
Beyond C	Between F and C	Real, inverted, diminished
At C	At C	Real, inverted, same size
Between F and C	Beyond C	Real, inverted, magnified
At F	At infinity	Real, inverted, highly magnified
Between P and F	Behind the mirror	Virtual, erect, magnified

Notes on the cases:

- **At infinity:** the incident rays are parallel and converge at F.
- **At F:** the two reflected rays come out parallel, so the image is at infinity. This is the reverse of the first case: object at infinity gives image at F, and object at F gives image at infinity. For the second ray you can use the ray through C, or the oblique ray at P. For a **point object** at F the reflected beam is parallel.
- **Between P and F:** this is the only case with a **virtual** image. The reflected rays **diverge**, so they are produced backwards (dashed) and *appear* to meet behind the mirror at **A'B'**. When using the oblique ray at P, make the two angles look equal.

All the real images are inverted and the virtual one is erect. Move about in front of a concave mirror and you see every one of these images in turn.

Uses of concave mirrors

- **Dentist's mirror** — the tooth sits between P and F, so the dentist sees an erect, magnified image.

- **Shaving and dressing-table mirrors** — your face is between P and F, so you see yourself erect and magnified.
- **Searchlights and torches** — a bulb at the **focus** of a concave reflector gives a parallel beam (the "object at $F \rightarrow$ image at infinity" case). A large area is lit over a long distance. The shiny aluminium-coated cup behind a torch bulb is this reflector.

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Convex Mirror Images, Sign Convention and the Mirror Formula

NCERT sections covered: 9.2.1, 9.2.3

Images formed by a convex mirror (NCERT 9.2.3)

A convex mirror has only two cases.

Position of object	Position of image	Nature and size
At infinity	At F, behind the mirror	Virtual, erect, highly diminished
Anywhere between infinity and P	Between P and F, behind the mirror	Virtual, erect, diminished

The parallel rays from an object at infinity reflect so that they *appear* to come from F. Draw the backward extensions dashed. Wherever the object is, a convex mirror's image is always **virtual, erect and smaller** than the object. (In the recording the teacher twice says "convex lens" while drawing the mirror, and once calls the outside car mirror "concave". The board and the argument are about a **convex mirror** both times.)

Why convex mirrors are used as rear-view mirrors

This is a common exam question. The mirror outside the driver's door bulges outwards, so it is a convex mirror. There are two reasons:

1. **The image is always erect.** The driver never sees a car or a person upside down.
2. **It gives a wider field of view.** Because it curves outwards, rays from objects at large angles behind the car still reflect into the driver's eye. The images are small, so a **large area behind is covered** in one small mirror.

With a plane mirror only rays within a narrow angle reach the eye, so the field of view is smaller. A concave mirror narrows it further. The convex mirror gives the widest view. (*Rear-view mirrors are not discussed in the current NCERT Class 12 text.*)

New Cartesian sign convention (NCERT 9.2.1)

For an object AB in front of a mirror with image A'B':

- **object distance** $u = PA$ (distance of the object from the pole)
- **image distance** $v = PA'$
- **focal length** $f = PF$, **radius of curvature** $R = PC$

Rules

1. **All distances are measured from the pole.** Write PA, PA', PF, never AP, A'P or FP.
2. Distances measured **in the direction of the incident ray are positive**. Distances measured **opposite to it are negative**. Draw the incident light travelling left to right.
3. Heights measured **above the principal axis are positive**, and heights **below are negative**.

	u	v	f	R
Concave mirror (real image)	–	–	–	–
Convex mirror	–	+	+	+

The object height h (AB, measured upwards) is positive. An inverted real image A'B' is below the axis, so its height is negative. The erect virtual image of a convex mirror has a positive height.

Memory hook from the lecture: "convex" gives a **positive** focal length, and "concave" a **negative** one. This holds for lenses too. The hook is only for the sign of f .

Mirror formula and magnification (NCERT 9.2.3)

The **mirror formula** relates the image distance, object distance and focal length:

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

Magnification m is the ratio of the image height to the object height:

$$m = \frac{h_i}{h_o} = -\frac{v}{u}$$

(Both are to be remembered here; they are derived in lecture 09.)

Reading m : the **sign** tells you real or virtual, and the **magnitude** tells you the size.

- $m > 0$: virtual and erect. $m < 0$: real and inverted.
- $|m| < 1$: smaller. $|m| = 1$: same size. $|m| > 1$: magnified.

m	Image
+2	magnified, erect, virtual
+0.2	smaller, erect, virtual
–3	magnified, real, inverted
–0.3	smaller, real, inverted

A common mistake is to read "+" as magnified and "-" as diminished. The sign only says virtual/erect or real/inverted.

Numericals (textbook exercise questions 12-15, solved on the board)

Q12. An object is 10 cm from a convex mirror of focal length 15 cm. Find the position and nature of the image. $u = -10$ cm, $f = +15$ cm.

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{15} + \frac{1}{10} = \frac{1}{6} \Rightarrow v = +6 \text{ cm}$$

The image is 6 cm **behind** the mirror, so it is virtual and erect. $m = -v/u = -\frac{6}{-10} = +0.6$: virtual, erect, smaller. (The teacher first writes $m = -0.6$, then notices that u is -10 and corrects it to $+0.6$.)

Q13. A plane mirror's magnification is $+1$. The "+" means virtual and erect, and the "1" means the same size.

Q14. An object 5.0 cm long is 20 cm in front of a convex mirror of radius of curvature 30 cm. $h_o = +5$ cm, $u = -20$ cm, $R = +30$ cm, so $f = R/2 = +15$ cm.

$$\frac{1}{v} = \frac{1}{15} + \frac{1}{20} = \frac{7}{60} \Rightarrow v = +\frac{60}{7} \approx +8.6 \text{ cm}$$

The image is behind the mirror, so it is virtual and erect. $m = -v/u = +\frac{3}{7}$, so it is smaller. Its height is $h_i = m h_o = \frac{15}{7} \approx 2.1$ cm.

Q15. An object 7.0 cm tall is 27 cm in front of a concave mirror of focal length 18 cm. Where should the screen be placed for a sharp image? $h_o = +7$ cm, $u = -27$ cm, $f = -18$ cm.

$$\frac{1}{v} = -\frac{1}{18} + \frac{1}{27} = \frac{18 - 27}{27 \times 18} = -\frac{1}{54} \Rightarrow v = -54 \text{ cm}$$

The negative v means the image is **in front of** the mirror, so it is real and inverted. Place the screen 54 cm in front of the mirror. $m = -v/u = -2$: real, inverted, magnified. $h_i = m h_o = -14$ cm (the minus means inverted).

Rule of thumb from the lecture: an image formed in front of the mirror is real and inverted, and one formed behind it is virtual and erect.

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Verify these spans

- [08:00-08:45] Segment times collapse here (several stamped 08:12) in the plane- vs concave-mirror field-of-view comparison; the text reads in order but its times are rough.

Refraction, Refractive Index and the Glass Slab

NCERT sections covered: 9.3

Refraction (NCERT 9.3)

You walk at different speeds on concrete, in a swimming pool and on sand dunes, because your speed depends on the medium. Light behaves the same way: **its speed depends on the medium.**

Definition (board): when light travels obliquely from one medium to another, it **bends at the interface**. This is called **refraction of light**. The *interface* is the surface separating the two media, for example the air-water boundary.

The bending happens because light travels at different speeds in air and water. If the speeds were equal it would go straight on. A ray that strikes the interface **normally** does not bend, but refraction still occurs, because its speed still changes.

Everyday examples

- A rod dipped in a glass of water looks **bent** at the water surface.
- **Coin in a beaker:** place your eye where the coin is just hidden. When a friend pours in water, the coin comes into view. You now see its **apparent position**, which is raised above its **real position**.
- A swimming pool looks shallower than it is, because its floor appears raised by refraction.
- Twinkling of stars and seeing the Sun for about 4 minutes more each day (advance sunrise, delayed sunset) are also due to refraction (taken up in lecture 10).

Optically denser and rarer media

- **Optically denser** medium: light travels **slower** in it. **Optically rarer:** light travels **faster**.
- Air and water: light is slower in water, so **air is optically rarer than water**.
- Water and glass: $v_{\text{water}} > v_{\text{glass}}$, so **water is optically rarer than glass**.

Optical density is not mass density. Oil floats on water: its density is 0.8 g cm^{-3} against water's 1 g cm^{-3} (mercury's is 13.6 g cm^{-3}). Yet oil is **optically denser** than water: $n_{\text{oil}} \approx 1.4 > n_{\text{water}} = 1.33$.

Summary (board): if a medium is optically denser, its refractive index is **more** and the speed of light in it is **less**. If it is optically rarer, its refractive index is **less** and the speed is **more**.

Just as densities are quoted relative to water, refractive indices are quoted relative to **air (or vacuum), taken as 1**. "The refractive index of glass is 1.5" means 1.5 with respect to air. For water it is 1.33 ($= 4/3$).

Refractive index measures how much light bends. For the same angle of incidence, light entering glass from air bends **more** than light entering water from air, so $n_{\text{glass}} > n_{\text{water}}$.

Direction of bending

- **Rarer $\rightarrow$ denser** (e.g. air $\rightarrow$ water): the ray bends **towards the normal**.
- **Denser $\rightarrow$ rarer** (e.g. water $\rightarrow$ air): the ray bends **away from the normal**.

- **Water** ($n_1 = 1.33$) $\rightarrow$ **oil** ($n_2 = 1.4$): oil is optically denser, so the ray bends **towards** the normal, even though oil is the lighter liquid.

Absolute and relative refractive index

The refractive index of a medium with respect to air is its **absolute refractive index**:

$$n_g = n_{ga} = \frac{n_g}{n_a} = \frac{1.5}{1} = 1.5, \quad n_w = n_{wa} = \frac{n_w}{n_a} = \frac{1.33}{1}$$

For two media neither of which is air, divide the absolute indices:

$$n_{wg} = \frac{n_w}{n_g} = \frac{n_w/n_a}{n_g/n_a} = \frac{1.33}{1.5}, \quad n_{gw} = \frac{n_g}{n_w} = \frac{1.5}{1.33}$$

Example. The refractive index of diamond is 2.54 and that of glass is 1.5. What is the refractive index of glass with respect to diamond?

$$n_{gd} = \frac{n_g}{n_d} = \frac{n_g/n_a}{n_d/n_a} = \frac{1.5}{2.54} \approx 0.59$$

Refraction through a rectangular glass slab

A ray in air meets the top face XY of a glass slab at angle of incidence i . Glass is denser, so the ray bends **towards** the normal; r is the **angle of refraction**. Inside the slab it travels straight, because bending happens only at an interface. At the bottom face AB it goes from denser to rarer and bends **away** from the normal, leaving at the **angle of emergence** e . The angle inside at AB is again r (alternate angles between the parallel faces).

- The bendings at XY and AB are **equal and opposite**, so $\angle i = \angle e$. The **emergent ray is parallel to the incident ray**.
- The emergent ray is shifted sideways. **Lateral displacement** d is the perpendicular distance between the (original) incident ray, produced forward, and the emergent ray.
- $d \propto t$, the **thickness** of the slab. Thickness is the distance between the incident face and the emergent face, so it depends on how the slab is placed. On the board, the slab standing on its long side has thickness $t_2 > t_1$, and its displacement (labelled d_1) is larger: $d_1 > d_2$.

Laws of refraction (NCERT 9.3)

For light going from medium 1 (index n_1) to medium 2 (index n_2):

1. **Snell's law.** The ratio of the sine of the angle of incidence to the sine of the angle of refraction is a constant, equal to the refractive index of medium 2 with respect to medium 1:

$$\frac{\sin i}{\sin r} = n_{21} = \frac{n_2}{n_1}$$

2. The incident ray, the refracted ray and the normal **all lie in the same plane**.

Hint to avoid mixing up n_1/n_2 : pair each angle with the index of the medium it lies in:

$$n_1 \sin i = n_2 \sin r$$

- Ray from n_4 into n_3 : $n_4 \sin i = n_3 \sin r$, so $\frac{\sin i}{\sin r} = \frac{n_3}{n_4}$.
- Ray from n_2 into n_1 : $n_2 \sin i = n_1 \sin r$, so $\frac{\sin i}{\sin r} = \frac{n_1}{n_2}$.

For the glass slab (air n_1 , glass n_2):

- At XY: $n_1 \sin i = n_2 \sin r$, so $\frac{\sin i}{\sin r} = \frac{n_2}{n_1}$.
- At AB: $n_2 \sin r = n_1 \sin e$, so $\frac{\sin r}{\sin e} = \frac{n_1}{n_2}$.

Since $e = i$, the second is just the first one inverted.

Refractive index from the speed of light

$$n = \frac{\text{speed of light in vacuum (or air)}}{\text{speed of light in the medium}}, \quad n_{ga} = \frac{c}{v} \quad (\text{absolute refractive index})$$

For any two media, the refractive index of medium 2 with respect to medium 1 is the speed in medium 1 divided by the speed in medium 2:

$$n_{21} = \frac{n_2}{n_1} = \frac{v_1}{v_2}$$

and the reverse, $n_{12} = n_1/n_2 = v_2/v_1$. The "master formula" from the lecture covers every case:

$$n_1 v_1 = n_2 v_2$$

Q. The speed of light is $2.5 \times 10^8 \text{ m s}^{-1}$ in medium 2 and $2 \times 10^8 \text{ m s}^{-1}$ in medium 3. If $n_2 = 1.5$, find n_3 .

$$n_2 v_2 = n_3 v_3 \Rightarrow n_3 = \frac{1.5 \times 2.5 \times 10^8}{2 \times 10^8} = 1.875$$

(The lecture stops at $1.5 \times 2.5/2$ and leaves the arithmetic to the class.)

Q. The speed of light in air is $3 \times 10^8 \text{ m s}^{-1}$. Find its speed in water, with $n_w = 4/3$.

$$1 \times 3 \times 10^8 = \frac{4}{3} v_2 \Rightarrow v_2 = \frac{9 \times 10^8}{4} = 2.25 \times 10^8 \text{ m s}^{-1}$$

(The board's last line reads "= 2 "; that is a slip, and the correct value is $2.25 \times 10^8 \text{ m s}^{-1}$.)

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30 s was re-transcribed on its own to confirm whether speech was missing. Part of the diamond numerical (about 23:25–24:06) was missing from the merged windows and was spliced in from a separate transcription; one sentence there appears twice in the transcript. Equations are taken from the board frames, not from the transcript.

Verify these spans

- [09:39–10:19] The transcript has the teacher say 'if velocity is more, then your refractive index will be more', followed by a single 34 s segment. That contradicts the board and the summary at 10:34 (higher speed means a lower index); either a slip of the tongue or a garbled stretch.

Lenses: Types and Terms

NCERT sections covered: 9.5.2

What is a lens? (NCERT 9.5.2)

Lens (board): a transparent material bounded by two surfaces, of which one or both are spherical, forms a lens. At least one surface has to be spherical: with both surfaces plane you get a slab, not a lens.

Convex (converging) lens

- **Both surfaces spherical and bulging outwards.** It is **thicker at the centre** than at the edges.
- Called a **double convex lens**, or simply a **convex lens**. If one face is plane, it is a **plano-convex** lens.
- Parallel rays falling on it **converge** to a point (F) after refraction, so a convex lens is a **converging lens**.

Concave (diverging) lens

- **Both surfaces spherical and curved inwards.** It is **thicker at the edges** than at the centre.
- Called a **double concave lens**, or simply a **concave lens**. If one face is plane, it is a **plano-concave** lens.
- Parallel rays falling on it **diverge** after refraction, as if coming from one point (F), so a concave lens is a **diverging lens**.

Meniscus lenses (an extra from the lecture)

When one face bulges out and the other curves in:

- thicker at the centre: **concavo-convex** lens;
- thinner at the centre: **convexo-concave** lens.

(The teacher notes these are not in the textbook.)

Terms related to a lens

- **Centres of curvature (C_1, C_2).** Each of a lens's two spherical surfaces is part of a sphere. The centres of those spheres are C_1 and C_2 .
- **Radii of curvature (R_1, R_2).** The radii of those two spheres. (*The teacher says this definition isn't spelled out in the textbook, so it is built from the C_1/C_2 one.*)
- **Principal axis.** The line passing through the two centres of curvature, C_1 and C_2 . Compare a mirror, which has only one spherical surface and so one C.
- **Optical centre (O).** The central point of a lens. **A ray passing through the optical centre goes undeviated.**
 - This holds only for a **thin lens**. Slice a thin lens into pieces: the central piece is effectively a very thin parallel-sided slab. A slab only displaces a ray sideways, and with negligible thickness the lateral shift is negligible, so the ray appears to pass straight through.
- **Aperture.** The effective diameter of the circular outline of a spherical lens (MN on the board). For a concave lens it is measured across the rim, not the thin middle.
- **Thin lens.** A lens whose aperture is much smaller than its radius of curvature. All the lens results here assume thin lenses.

Principal focus and focal length

Convex lens. When several rays parallel to the principal axis fall on a convex lens, after refraction they **converge** at a point on the principal axis. This point is the **principal focus** of the lens.

Concave lens. When several rays parallel to the principal axis fall on a concave lens, after refraction they **appear to diverge** from a point on the principal axis. This point is the **principal focus**.

Focal length (f). The distance between the optical centre and the principal focus.

Drawing convention: for a thin lens, draw the bending of each ray at the **centre line** of the lens, not at the two surfaces separately.

Two principal foci. If parallel rays come from the other side of the lens, they focus on the opposite side. So a lens has **two principal foci**, F_1 and F_2 ("focus" is singular, "foci" plural). This answers "why does a lens have two principal foci?"

- $OF_1 = OF_2$ **only if the medium on both sides of the lens is the same**, for example air | glass ($n = 1.5$) | air.
- With air on one side and water on the other, the two focal lengths differ.

(This last point is an extra from the lecture. In the course, both sides are always the same medium.)

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true length, so times quoted here are approximate (roughly ± 30 s). Every gap of more than 30 s was re-transcribed on its own to confirm whether speech was missing. Equations are taken from the board frames, not from the transcript.

Lens Ray Diagrams, Lens Formula and Power

NCERT sections covered: 9.5.2, 9.5.3, 9.5.4

Important rays for lens ray diagrams (NCERT 9.5.2)

1. A ray parallel to the principal axis

- **Convex lens:** after refraction it passes through the focus **on the other side** of the lens.
- **Concave lens:** it *appears* to pass through (diverge from) the focus **on the same side**.

2. A ray through the focus becomes parallel to the principal axis after refraction.

- For a concave lens, this is the ray **directed towards** the focus on the far side.

3. A ray through the optical centre goes undeviated, for both lenses.

For a thin lens, show the bending at the centre line of the lens. Two rays locate the image.

Images formed by a convex lens (NCERT 9.5.2)

Mark O, F and 2F on **both** sides of the lens. For a lens, 2F plays the role that C played for a mirror.

Position of object	Position of image	Nature and size
At infinity	At F, on the other side	Real, inverted, highly diminished
Beyond 2F	Between F and 2F	Real, inverted, diminished
At 2F	At 2F	Real, inverted, same size
Between F and 2F	Beyond 2F	Real, inverted, magnified
At F	At infinity	Real, inverted, highly magnified
Between O and F	On the same side as the object (in front of the lens)	Virtual, erect, magnified

- **At F:** a **point object** gives a parallel beam. For an **extended object**, the parallel ray (through F') and the ray through O come out parallel, so the image is at infinity.
- **Between O and F:** the two refracted rays diverge. Produced backwards, they meet on the object's side. This is the only case where the image is virtual and on the same side as the object. It is the principle of the **simple microscope** (magnifying glass), the lens a palm-reader holds over your hand. The object must lie between O and F.
- Drawing tips are the same as for mirrors: for the "beyond 2F" case draw the object a little larger, and for "at 2F" mark A'B' first.

Images formed by a concave lens

Position of object	Position of image	Nature and size
At infinity	Appears to be at F (same side)	Virtual, erect, highly diminished
Anywhere between infinity and O	Between O and F, same side as the object	Virtual, erect, diminished

Sign convention for lenses (NCERT 9.5)

1. All distances are measured from the optical centre O: $u = OA$, $v = OA'$, $f = OF$.
2. Distances **in the direction of the incident light are positive**; distances against it are negative.
3. Heights **above the principal axis are positive**; heights below it are negative.

	u	v	f
Convex lens (real image)	–	+	+
Concave lens	–	–	–

- A convex lens converges rays to the focus on the far side, in the direction of the incident light, so $f > 0$.
- A concave lens's focus is where the rays *appear* to come from, on the incident side, so $f < 0$.
- This fits the "convex $\rightarrow$ positive f , concave $\rightarrow$ negative f " hook from lecture 05.
- Heights: the inverted real image of a convex lens has a negative height. Both AB and $A'B'$ are positive for a concave lens.

Lens formula and magnification (NCERT 9.5.2)

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}, \quad m = \frac{h_i}{h_o} = \frac{v}{u}$$

Compare the mirror versions, $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$ and $m = -\frac{v}{u}$.

The sign of m gives the nature: + means virtual and erect, – means real and inverted. Its magnitude gives the size: $|h_i| < h_o$ is smaller, = is the same size, > is magnified. For example, $m = +2$ is virtual, erect and magnified; $m = +0.2$ is virtual, erect and smaller; $m = -5$ is real, inverted and magnified.

Power of a lens (NCERT 9.5.3)

Send parallel rays into two convex lenses. Lens 1 bends the ray through a larger angle ($\theta > \theta'$), so its **degree of convergence** is greater. We say lens 1 has the greater **power**. The bigger bend brings the ray to the axis sooner, so its focal length is shorter ($f_1 < f_2$).

Power is the reciprocal of focal length:

$$P = \frac{1}{f} \quad (f \text{ in metres})$$

The SI unit of power is the **diopetre (D)**; $1 \text{ D} = 1 \text{ m}^{-1}$.

- $f > 0$ gives $P > 0$: a **convex** lens.
- $f < 0$ gives $P < 0$: a **concave** lens. A lens of power -1 D is concave.
- **Always convert f to metres** before taking the reciprocal. Forgetting is the most common mistake.

Example. Find the power of a concave lens of focal length 50 cm . $f = -50 \text{ cm} = -0.5 \text{ m}$, so $P = 1/f = -2 \text{ D}$.

Rule of thumb from the lecture: a thick spectacle lens has a higher power and a shorter focal length, and a thin one has a lower power and a longer focal length. Strictly, the focal length depends on how curved the surfaces are; see the lens-maker formula in lecture 11.

Lenses in contact (NCERT 9.5.4)

For thin lenses in contact (here convex f_1 , convex f_2 , concave f_3):

$$\frac{1}{f_{\text{eff}}} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3}, \quad P_{\text{eff}} = P_1 + P_2 + P_3$$

Example: $f_1 = 5 \text{ cm}$, $f_2 = 10 \text{ cm}$, $f_3 = -2 \text{ cm}$.

$$\frac{1}{f_{\text{eff}}} = \frac{1}{5} + \frac{1}{10} - \frac{1}{2} = \frac{4 + 2 - 10}{20} = -\frac{4}{20} = -\frac{1}{5}$$

so $f_{\text{eff}} = -5 \text{ cm} = -0.05 \text{ m}$ and $P_{\text{eff}} = -20 \text{ D}$. The combination acts as a diverging lens.

Board slip: the board (and the recording) gives $4 + 2 - 10 = -2$, which leads to $f_{\text{eff}} = -10 \text{ cm}$ and $P = -10 \text{ D}$. The correct values are -5 cm and -20 D . Cross-check with powers: $+20 + 10 - 50 = -20 \text{ D}$.

Textbook questions solved in the lecture

Q9. One half of a convex lens is covered with black paper. Does it still form a complete image? Yes. Every part of the lens sends rays from every point of the object to the image. The two rays in a diagram are only the convenient ones for locating it. Covering half the lens removes half the rays, so the image is complete but its **intensity (brightness) is reduced, to about half**.

Q10. An object 5 cm tall is 25 cm from a converging lens of focal length 10 cm . ($2F = 20 \text{ cm}$, so the object is beyond $2F$.) $h_o = +5 \text{ cm}$, $u = -25 \text{ cm}$, $f = +10 \text{ cm}$.

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{10} - \frac{1}{25} = \frac{3}{50} \Rightarrow v = +\frac{50}{3} \approx 16.7 \text{ cm}$$

The image is on the other side: real and inverted. $m = v/u = -\frac{2}{3} \approx -0.67$, so it is smaller. $h_i = m h_o = -\frac{10}{3} \approx -3.3 \text{ cm}$.

Q11. A concave lens of focal length 15 cm forms an image 10 cm from the lens. How far is the object? $f = -15 \text{ cm}$, $v = -10 \text{ cm}$.

$$\frac{1}{u} = \frac{1}{v} - \frac{1}{f} = -\frac{1}{10} + \frac{1}{15} = -\frac{1}{30} \Rightarrow u = -30 \text{ cm}$$

The object is 30 cm in front of the lens.

Q16. Find the focal length of a lens of power -2.0 D . $f = 1/P = -0.5 \text{ m} = -50 \text{ cm}$, a diverging (concave) lens.

Q17. A doctor prescribes a corrective lens of power $+1.5 \text{ D}$. Find its focal length; is it converging or diverging? $f = 1/1.5 \text{ m} = 200/3 \approx +66.7 \text{ cm}$. Positive, so it is **converging** (convex).

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Derivations: $f = R/2$, Mirror Formula, Lens Formula, Lenses in Contact

NCERT sections covered: 9.2.2, 9.2.3, 9.5.2, 9.5.4

Four derivations, each with its assumption stated first. The lecture stresses writing the assumption in the exam.

1. Relation between f and R for a spherical mirror (NCERT 9.2.2)

Assumption: the aperture of the mirror is small.

Concave mirror.

1. A ray parallel to the principal axis strikes the mirror at M and reflects through F.
2. CM is the normal at M, so $\angle i = \angle r$.
3. $\angle FCM = i$ (alternate angles, since the incident ray is parallel to the axis), and $\angle FMC = r = i$.
4. So $\triangle CFM$ is **isosceles**, and $CF = FM$.
5. Because the aperture is small, M is very close to P, so $FM \approx FP$.
6. Hence $CF = FP$: **F is the midpoint of P and C**, and $R = PC = 2PF$:

$$R = 2f \quad \text{or} \quad f = \frac{R}{2}$$

Convex mirror. The same steps work behind the mirror:

- the normal at M is CM produced;

- the reflected ray produced backwards passes through F;
- $\triangle FMC$ is isosceles with $FM = FC$;
- small aperture gives $FP = FC$, so again $R = 2f$.

2. Mirror formula (NCERT 9.2.3)

Assumption: the aperture of the mirror is small.

Setup. Take a concave mirror with object AB beyond C.

- The ray from B parallel to the axis hits the mirror at M and reflects through F. The ray from B through F reflects parallel. They meet at B', so the image A'B' is between C and F.
- Drop MN perpendicular to the axis. Then $MN = AB$, and by the small aperture, $N \approx P$.
- Distances: $PA = u$, $PA' = v$, $PF = f$, so $FA' = PA' - PF = v - f$.

Triangles A'B'F and MNF are similar:

$$\frac{A'B'}{AB} = \frac{FA'}{PF} = \frac{v - f}{f} \quad (1)$$

Triangles ABP and A'B'P are similar (the ray to the pole reflects at equal angles):

$$\frac{A'B'}{AB} = \frac{PA'}{PA} = \frac{v}{u} \quad (2)$$

Magnification from (2). A'B' is below the axis, so $A'B' = -h_i$, while $PA' = -v$ and $PA = -u$:

$$\frac{-h_i}{h_o} = \frac{-v}{-u} \Rightarrow m = \frac{h_i}{h_o} = -\frac{v}{u}$$

Equating (1) and (2) and putting in the sign convention ($u \rightarrow -u$, $v \rightarrow -v$, $f \rightarrow -f$ for a concave mirror with a real image):

$$\frac{-v - (-f)}{-f} = \frac{-v}{-u} \Rightarrow \frac{-v + f}{-f} = \frac{v}{u}$$

Cross-multiplying gives $-uv + uf = -vf$. Dividing by uvf :

$$-\frac{1}{f} + \frac{1}{v} = -\frac{1}{u} \Rightarrow \boxed{\frac{1}{f} = \frac{1}{v} + \frac{1}{u}}$$

3. Lens formula (convex lens)

Assumption: the lens is thin. The ray through O then goes undeviated, and distances can be measured from O.

Setup. Object AB is between F and 2F, and the image A'B' is beyond 2F on the other side.

- The ray from B parallel to the axis meets the lens at P (so $OP = AB$) and refracts through F.
- The ray from B through O goes straight on.
- Distances: $OA = u$, $OA' = v$, $OF = f$, so $FA' = v - f$.

Triangles ABO and A'B'O are similar:

$$\frac{A'B'}{AB} = \frac{OA'}{OA} = \frac{v}{u} \quad (1)$$

Triangles POF and A'B'F are similar (with $OP = AB$):

$$\frac{A'B'}{AB} = \frac{FA'}{OF} = \frac{v-f}{f} \quad (2)$$

Magnification from (1). With $A'B' = -h_i$, $OA' = +v$ and $OA = -u$:

$$\frac{-h_i}{h_o} = \frac{v}{-u} \Rightarrow m = \frac{h_i}{h_o} = \frac{v}{u}$$

There is no minus sign here, unlike a mirror.

Equating (1) and (2) with signs ($OA = -u$, $OA' = +v$, $OF = +f$):

$$\frac{v}{-u} = \frac{v-f}{f} \Rightarrow vf = -uv + uf$$

Dividing by uvf :

$$\frac{1}{u} = -\frac{1}{f} + \frac{1}{v} \Rightarrow \boxed{\frac{1}{f} = \frac{1}{v} - \frac{1}{u}}$$

(When setting this up, the teacher at one point says " $OA' = -v$ "; the board and the working use $OA' = +v$. NCERT itself reaches the lens formula differently, by combining refraction at the two spherical surfaces — see lecture 11.)

4. Combination of thin lenses in contact (NCERT 9.5.4)

This derivation has been asked in board exams. **Assumption: the lenses are thin**, so their thickness and separation are negligible, and u and v can be measured from either lens.

Setup.

- Two lenses L_1 (focal length f_1) and L_2 (focal length f_2) are in contact.
- An object O is at distance u .
- Lens 1 alone would form an image I' at distance v' .
- I' acts as a **virtual object** for lens 2, which forms the final image I at distance v .

For lens 1:

$$\frac{1}{f_1} = \frac{1}{v'} - \frac{1}{u} \quad (1)$$

For lens 2:

$$\frac{1}{f_2} = \frac{1}{v} - \frac{1}{v'} \quad (2)$$

Adding (1) and (2), the $1/v'$ terms cancel:

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_1} + \frac{1}{f_2}$$

The pair behaves as a single lens that takes an object at u to an image at v . Its focal length f therefore satisfies $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, so

$$\boxed{\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}} \quad \text{and, since } P = \frac{1}{f}, \quad \boxed{P = P_1 + P_2}$$

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Lateral Displacement, Apparent Depth and Atmospheric Refraction

NCERT sections covered: 9.3

Lateral displacement in a glass slab (NCERT 9.3)

Setup and what happens to the ray.

- A ray enters a slab of thickness t at O, with angle of incidence i and angle of refraction r .
- It leaves at B. The angle inside at the lower face is again r , and the ray emerges at angle e .
- The incident and emergent rays are **parallel**, so $i = e$.
- The ray is shifted sideways. The **lateral displacement** $d = BN$ is the perpendicular distance from B to the incident ray produced.

Thickness is the distance between the face where the ray enters and the face where it leaves. Turn the slab and the thickness changes with it.

Derivation (board). The incident ray, produced forward, makes the angle $i - r$ with the refracted ray OB.

In right $\triangle OBN$:

$$\sin(i - r) = \frac{BN}{OB} = \frac{d}{OB} \quad (1)$$

In right $\triangle ON'B$, where N' is the foot of the normal at the lower face:

$$\cos r = \frac{t}{OB} \quad (2)$$

Dividing (1) by (2) cancels OB:

$$d = \frac{t \sin(i - r)}{\cos r}$$

For a given i and slab material, $d \propto t$: the thicker the slab the ray crosses, the larger the lateral displacement. (NCERT shows the lateral shift only qualitatively, in Fig. 9.9. This expression is an extra from the lecture.)

Real and apparent depth (NCERT 9.3)

A coin at the bottom of a water-filled container appears **raised**. Rays from the coin go from water (denser) into air (rarer) and bend **away from the normal**. Traced back, they seem to come from a point above the coin.

Let h_1 be the **real depth** and h_2 the **apparent depth**. For **normal viewing**, with the eye close to the surface and looking nearly straight down:

$$\frac{n_2}{n_1} = \frac{h_1}{h_2} = \frac{\text{real depth}}{\text{apparent depth}} \Rightarrow n = \frac{h_{\text{real}}}{h_{\text{app}}}$$

Here $n = n_w/n_a = 1.33$. Since $n > 1$, $h_1 > h_2$. Viewed obliquely, from far off, the calculation is different.

Shift (how far the coin appears to rise):

$$\Delta h = h_1 - h_2 = h_1 - \frac{h_1}{n} = h_1 \left(1 - \frac{1}{n}\right)$$

(NCERT names them the other way round, with h_1 apparent and h_2 real. The relation "apparent depth = real depth $\div$ n , for near-normal viewing" is the same.)

The observer inside the water: a fish looking at a fisherman

Now the object (the fisherman) is in air and the observer (the fish) is in water. Light from the fisherman goes from rarer to denser, bending **towards** the normal. Traced back, it seems to come from higher up, so **to a fish, the fisherman appears raised (taller)**.

Master formula (board):

$$d' = \frac{d}{n_{\text{medium}}} \times n_{\text{observer}}$$

Here d' is the apparent distance and d the real distance. n_{medium} is the refractive index where the object is, and n_{observer} where the eye is.

- **Coin, observer in air:** $d' = d \times 1/n$, which is the result above.
- **Fisherman, fish in water (n_2):** $d' = d \times n_2/1$, so $n = \frac{d'}{d} = \frac{h_{\text{app}}}{h_{\text{real}}}$. This is the opposite ratio to the coin case. The master formula handles both without having to remember which is which.

Several layers. A coin lies under liquid layers of thicknesses d_2, d_3, d_4 and indices n_2, n_3, n_4 , with the observer in air just above the surface:

$$d' = 1 \left(\frac{d_2}{n_2} + \frac{d_3}{n_3} + \frac{d_4}{n_4} \right)$$

If the observer is a height d_5 above the surface, add $d_5/1$ (air). The result is then measured from the eye.

Shift caused by a glass slab

An object is viewed through a slab of thickness t and index n (relative to the surroundings, $n = n_2/n_1$):

1. The first face forms an image O' .
2. O' acts as the object for the second face, which forms the final image O'' .
3. The observer sees the object at O'' (O' is only an intermediate step).

$$\text{shift} = t \left(1 - \frac{1}{n} \right)$$

The shift is **in the direction of the incident light** (towards the observer).

Example. With $t = 1$ cm and $n = 1.5$:

$$\text{shift} = 1 \left(1 - \frac{1}{1.5} \right) = \frac{0.5}{1.5} = \frac{1}{3} \text{ cm}$$

An object 5 cm away appears $5 - \frac{1}{3}$ cm away.

Board-exam question (3 marks)

(a) A convex lens forms an image at $v = 10$ cm. A glass slab ($t = 3$ cm, $n = 1.5$) is placed between the lens and the image. Where is the image now?

$$\text{shift} = 3 \left(1 - \frac{1}{1.5} \right) = 1 \text{ cm}$$

The shift is along the light, away from the lens. So $v = 10 + 1 = 11$ cm, and the screen must be moved to 11 cm.

(b) The same slab is placed between an object (10 cm from the lens) and the lens. The object appears shifted 1 cm towards the lens, to O' . So the lens "sees" $u = 10 - 1 = 9$ cm, and v follows from the lens formula with that u .

Atmospheric refraction (not in the current NCERT text)

Refraction by the Earth's atmosphere. The air is denser, with a higher refractive index, near the ground and rarer higher up.

1. Advance sunrise and delayed sunset (by about 2 minutes each).

- When the Sun is just below the horizon, its rays enter the atmosphere going from rarer to denser layers and bend **towards the normal**, curving round towards the Earth.

- An observer sees the Sun at its **apparent position**, above the horizon, while its **real position** is still below it.
- The apparent shift is about $\frac{1}{2}^\circ$. The Earth turns 360° in 24 h, so $\frac{1}{2}^\circ$ takes 2 minutes.
- The Sun is therefore seen 2 min before actual sunrise and 2 min after actual sunset: **about 4 minutes more daylight each day**.

2. Twinkling of stars.

- Starlight passes through layers of increasing refractive index on its way down and bends towards the normal again and again. The star is seen at an **apparent position slightly higher** than its real position.
- **The atmosphere is dynamic.** Its layers move, so their density and refractive index keep changing slightly, and the path of the star's light keeps changing.
- **Board text:** "Since atmosphere is dynamic it results in changes in density & refractive index of layers. This causes a change in path of rays coming from the star as they pass through them. Thus sometimes more rays fall upon the eye of the observer & star appears brighter, & sometimes only few rays reach the eye & it appears fainter. Thus star appears twinkling."

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Refraction at Spherical Surfaces and the Lens-Maker's Formula

NCERT sections covered: 9.5.1, 9.5.2

Refraction at a spherical surface (NCERT 9.5.1)

Assumptions. In optics these must be stated before any derivation:

1. The object is a **point object on the principal axis**.
2. The angles made by the incident and refracted rays with the principal axis are **small**.
3. The **aperture is small**.

(a) Convex spherical surface, light going from n_1 (rarer) to n_2 (denser)

Setup.

- A ray from the point object O strikes the surface at N.
- CN is the normal (C is the centre of curvature). The ray bends towards the normal and meets the axis at the image I.
- Drop NP' perpendicular to the axis. With a small aperture, P' coincides with the pole P.
- Angles with the axis: α for the incident ray, β for the refracted ray, γ for the normal.

- Distances: $PO = u$, $PI = v$, $PC = R$.

Exterior-angle property (exterior angle = sum of the interior opposite angles):

$$\triangle ONC : i = \alpha + \gamma \quad \triangle NCI : \gamma = r + \beta \Rightarrow r = \gamma - \beta$$

Snell's law at N. The rule "each angle goes with the index of its own medium" gives $n_1 \sin i = n_2 \sin r$. For small angles $\sin i \approx i$ and $\sin r \approx r$:

$$n_1(\alpha + \gamma) = n_2(\gamma - \beta)$$

Small angles, so each angle $\approx$ its tangent:

$$n_1(\tan \alpha + \tan \gamma) = n_2(\tan \gamma - \tan \beta)$$

$$n_1 \left(\frac{NP}{-u} + \frac{NP}{R} \right) = n_2 \left(\frac{NP}{R} - \frac{NP}{v} \right)$$

The signs used are $u < 0$, $R > 0$, $v > 0$. Cancel NP and rearrange:

$$\boxed{\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}}$$

Memory hint. Start from the lens formula $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$:

- put the index of the medium each distance lies in on top: v is in n_2 and u is in n_1 ;
- the right side is (index you are going **to** – index you are coming **from**) over R .

Numericals are set on this formula.

(b) Concave spherical surface, light going from n_2 (denser) to n_1 (rarer)

The assumptions are the same. The object O is in n_2 , with the centre of curvature on the same side. The ray bends **away** from the normal and forms a real image I in n_1 .

Angle relations:

$$\triangle ONC : \alpha + i = \gamma \Rightarrow i = \gamma - \alpha \quad \triangle CNI : r = \gamma + \beta$$

Snell's law, then the small-angle step: $n_2 \sin i = n_1 \sin r$ becomes $n_2 i = n_1 r$. Then

$$n_2 \left(\frac{NP}{-R} - \frac{NP}{-u} \right) = n_1 \left(\frac{NP}{-R} + \frac{NP}{v} \right)$$

The signs are $R < 0$ (C lies against the incident light), $u < 0$, $v > 0$. This gives:

$$\boxed{\frac{n_1}{v} - \frac{n_2}{u} = \frac{n_1 - n_2}{R}}$$

The result is the same formula with $n_1 \leftrightarrow n_2$ swapped, because the light now goes from n_2 to n_1 . Whether the image is real or virtual, the equation is unchanged.

Lens-maker's formula (NCERT 9.5.2)

Assumptions: the same as above, **plus the lens is thin**. All distances can then be measured from the optical centre O, since P_1 and P_2 almost coincide.

Setup. A lens of index n_2 sits in a medium of index n_1 . Its first surface LP_1M has radius R_1 , and its second surface LP_2M has radius R_2 .

Step 1: refraction at LP_1M . Pretend the lens material fills all the space to the right. The first surface alone forms an image I' at distance v' , inside n_2 :

$$\frac{n_2}{v'} - \frac{n_1}{u} = \frac{n_2 - n_1}{R_1} \quad (1)$$

Step 2: refraction at LP_2M . I' acts as a **virtual object** for the second surface (it lies in n_2). The final image I is at v , in n_1 :

$$\frac{n_1}{v} - \frac{n_2}{v'} = \frac{n_1 - n_2}{R_2} \quad (2)$$

Adding (1) and (2). The n_2/v' terms cancel:

$$\frac{n_1}{v} - \frac{n_1}{u} = (n_2 - n_1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\frac{1}{v} - \frac{1}{u} = \frac{n_2 - n_1}{n_1} \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

The left side is $1/f$ (lens formula), so:

$$\boxed{\frac{1}{f} = \frac{n_2 - n_1}{n_1} \left(\frac{1}{R_1} - \frac{1}{R_2} \right)}$$

This is the **lens-maker's formula**, used to design lenses of a desired focal length. Here n_1 is the index of the **surrounding medium**. In NCERT's notation, $\frac{n_2 - n_1}{n_1} = n_{21} - 1$.

Note: when you substitute signed radii, do not cancel $\frac{1}{R_1}$ against $\frac{1}{R_2}$. For a biconvex lens R_2 is negative, so the two terms add.

Can a convex lens behave as a diverging lens?

Take a convex lens of glass, $n_2 = 1.5$. For a convex lens, $\left(\frac{1}{R_1} - \frac{1}{R_2} \right) > 0$.

1. $n_2 > n_1$ (in air, $n_1 = 1$):

$$\frac{1}{f} = \frac{1.5 - 1}{1} \left(\frac{1}{R_1} - \frac{1}{R_2} \right) > 0$$

f is positive: a **converging** lens, the normal case.

2. $n_2 = n_1$ (in a liquid of index 1.5):

$$\frac{1}{f} = \frac{1.5 - 1.5}{1.5}(\dots) = 0 \Rightarrow f = \infty$$

The rays go straight through: it behaves as **plane glass**.

3. $n_2 < n_1$ (in a liquid of index 1.6):

$$\frac{1}{f} = \frac{1.5 - 1.6}{1.6} \left(\frac{1}{R_1} - \frac{1}{R_2} \right) = -\frac{0.1}{1.6}(\dots)$$

f is negative: the convex lens behaves as a **diverging (concave) lens**.

Q. An air bubble inside water: does it behave as a convex or a concave lens? The bubble is the "lens" ($n_2 = 1$) and the water is the surroundings ($n_1 = 1.33$). It is bulging, so $(1/R_1 - 1/R_2) > 0$:

$$\frac{1}{f} = \frac{1 - 1.33}{1.33} \left(\frac{1}{R_1} - \frac{1}{R_2} \right) = -\frac{0.33}{1.33}(\dots) < 0$$

f is negative, so the air bubble behaves as a **concave (diverging) lens**.

Can a concave lens behave as a converging lens?

Case	Behaviour
$n_2 > n_1$ (normal)	diverging lens (concave)
$n_2 = n_1$	plane glass, $f = \infty$
$n_1 > n_2$ (e.g. lens 1.5 in liquid 1.6)	converging lens: both factors are negative, so $f > 0$

Lens vs mirror

A lens's focal length depends on the refractive indices through the factor $\frac{n_2 - n_1}{n_1}$. It can be positive, negative or infinite depending on the medium the lens is in. Put a lens in water and its focal length changes.

A mirror's focal length, $f = R/2$, has **no refractive index in it**. A mirror's focal length is the same in water as in air.

Note on this lecture's transcript: gemini-3.5-flash was in a 503 outage and then out of daily quota when this chapter was processed, so the audio was transcribed by gemini-3.5-flash-lite in overlapping five-minute windows, joined where the overlapping speech matches word for word. Lite's clock ran fast (by up to 2x) inside a window; each window was rescaled to its true length, so times quoted here are approximate (roughly ± 30 s). Every gap of more than 30 s was re-transcribed on its own to confirm whether speech was missing. The start of the concave-surface derivation (about 11:26-12:02) was partly missing from the merged windows and was spliced in from a separate transcription; a few sentences there appear twice in the transcript. Equations are taken from the board frames, not from the transcript.

Prism, Dispersion, Rainbow and Scattering

NCERT sections covered: 9.6

Prism and dispersion

Prism (board). A homogeneous transparent medium, such as glass, enclosed by two plane refracting surfaces is called a prism. A glass prism has **3 rectangular faces and 2 triangular faces**.

Dispersion (board). The splitting of white light into its constituent colours when it passes through a prism. The colours spread out as **VIBGYOR**, with **violet deviated the most and red the least**: $\delta_V > \delta_R$.

Why does dispersion happen? The refractive index of a medium is **different for different colours**. " $n_g = 1.5$ " is only a typical value. Cauchy's formula gives

$$n = A + \frac{B}{\lambda^2} + \frac{C}{\lambda^4} + \dots \quad (A, B, C \text{ constants})$$

so n decreases as wavelength increases (the lecture writes this loosely as $n \propto 1/\lambda$). Since $\lambda_R > \lambda_V$, we get $n_V > n_R$, and violet bends more. (*Current NCERT mentions dispersion only in the chapter summary. Cauchy's formula is an extra.*)

Rainbow (not in the current NCERT text)

A rainbow is natural dispersion.

Conditions:

- tiny water droplets must be suspended in the air, after rain or near a garden fountain; the droplets act as prisms;
- **the observer's back must face the Sun.**

Primary rainbow.

- Sunlight is refracted on entering a drop, **reflected once** inside it, and refracted again on leaving.
- Violet reaches the eye at about **40°** to the anti-solar direction and red at about **42°**, so the band is about 2° wide.
- The colours are in the usual order, **red on the outside and violet on the inside**.

Secondary rainbow.

- The light enters the lower part of the drop and is **reflected twice** inside it.
- The angles are about **50°-53°**, a band of about 3°, seen **above** the primary.
- Its **colours are reversed**: violet is on the outside.

(The lecture calls the reflections inside the drop "total internal reflection". For the rays that form the bow, the angle inside the drop, about 40°, is below water's critical angle of about 49°. So the reflection is really ordinary partial reflection, which is also why rainbows are faint.)

Usually we see only an **arc**, because the ground cuts off the rest of the circle. From an aircraft, a **full circular rainbow** can be seen.

Refraction through a prism (NCERT 9.6)

Setup.

- A ray strikes the first face at angle of incidence i and is refracted at r_1 .
- It meets the second face at r_2 and emerges at angle of emergence e .
- The deviation at the first face is $\delta_1 = i - r_1$, and at the second face $\delta_2 = e - r_2$.
- The total deviation δ is the exterior angle of the triangle formed by the incident ray produced and the emergent ray produced back, so

$$\delta = \delta_1 + \delta_2 = (i - r_1) + (e - r_2) = i + e - (r_1 + r_2)$$

Angle of prism A: the angle between the two refracting (rectangular) faces.

Showing $r_1 + r_2 = A$. Let N be the point where the two normals meet inside the prism.

- In the quadrilateral formed by the apex, the two points of incidence and N, the angles at the two points of incidence are 90° (each normal is perpendicular to its face). The angles sum to 360° , so

$$90^\circ + A + 90^\circ + \angle N = 360^\circ \Rightarrow A + \angle N = 180^\circ$$

- In the triangle formed by the two points of incidence and N, $r_1 + r_2 + \angle N = 180^\circ$.
- Comparing the two:

$$\boxed{r_1 + r_2 = A}$$

Substituting into the deviation:

$$\boxed{\delta = i + e - A} \quad \text{or} \quad \delta + A = i + e$$

Angle of minimum deviation

Measure δ for $i = 30^\circ, 35^\circ, 40^\circ, \dots$ and plot δ against i . The deviation **falls, reaches a minimum δ_m , then rises**.

At $\delta = \delta_m$:

1. $i = e$;
2. the ray inside the prism travels **parallel to the base**;
3. $r_1 = r_2 = r$, and since $r_1 + r_2 = A$, $2r = A$, so $r = A/2$.

Board slip: the board boxes this as " $A = r/2$ ". The correct relation is $r = A/2$, and the teacher corrects herself a minute later when using it.

Prism formula.

- From $i + e = A + \delta_m$ with $i = e$: $i = \frac{A + \delta_m}{2}$.
- Snell's law at the first face, $n_1 \sin i = n_2 \sin r$, then gives

$$\boxed{n_{21} = \frac{n_2}{n_1} = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}}$$

The 2s in the numerator and denominator **cannot be cancelled**, because each sits inside a sine.

Deviation by a thin prism (NCERT 9.6)

A prism with $A < 10^\circ$ is a **thin prism**, and all its angles are small, so $\sin \theta \approx \theta$.

Snell's law at the two faces: $n_1 i = n_2 r_1$ and $n_2 r_2 = n_1 e$. So, with $n = n_2/n_1$,

$$i = nr_1, \quad e = nr_2$$

Deviation:

$$\delta = i + e - A = n(r_1 + r_2) - A = nA - A \Rightarrow \boxed{\delta = (n - 1)A}$$

For a thin prism the deviation depends only on A and n . It is **independent of the angle of incidence**, whereas for a general prism δ depends on i and e .

Angular dispersion and dispersive power (not in the current NCERT text)

Angular dispersion. The angle between the emergent rays of two colours (violet and red):

$$\theta = \delta_V - \delta_R$$

For a thin prism:

$$\theta = (n_V - 1)A - (n_R - 1)A = (n_V - n_R)A$$

It depends on the angle of the prism.

Dispersive power (ω). The ratio of the angular dispersion to the **mean deviation**, taken as the deviation of yellow light:

$$\omega = \frac{\delta_V - \delta_R}{\delta_Y} = \frac{(n_V - n_R)A}{(n_Y - 1)A} = \frac{n_V - n_R}{n_Y - 1}$$

- ω is **independent of the angle of prism** and depends only on the material.
- A prism that spreads white light over a wider fan has the greater dispersive power.
- $n_{\text{flint}} > n_{\text{crown}}$, and $\omega_{\text{flint}} > \omega_{\text{crown}}$.
- If a numerical does not give n_Y , take $n_Y = \frac{n_V + n_R}{2}$.

Scattering of light (not in the current NCERT text)

What scattering is. Air molecules, dust and smoke particles in the atmosphere deflect light from its path. This happens through the interaction of the light wave's electric field with the particle. The deflected light is **scattered light**.

Rayleigh scattering. For particles **smaller than (or about equal to) the wavelength**, the scattered intensity is

$$I \propto \frac{1}{\lambda^4}$$

So fine particles scatter violet and blue far more strongly than red.

Blue sky. Blue and violet are scattered strongly all over the sky. Violet is partly lost on the way by further scattering, so the scattered skylight reaching us is rich in **blue**.

- **With no atmosphere, the sky would look black** even in daytime, because there would be nothing to scatter the light.

White clouds. Cloud droplets are **much larger than the wavelength**. Then all colours are scattered about equally, and the clouds look **white**.

Red Sun at sunrise and sunset (board). At sunrise and sunset, sunlight travels a **longer path** through the atmosphere before it reaches the observer's eye directly. Violet and blue are strongly scattered out on the way, so the light reaching the observer is deprived of violet and blue and mostly red reaches the eye. Thus the Sun at sunrise and sunset appears **red**.

Note on this lecture's transcript: gemini-3.5-flash was in a 503 outage and then out of daily quota when this chapter was processed, so the audio was transcribed by gemini-3.5-flash-lite in overlapping five-minute windows, joined where the overlapping speech matches word for word. Lite's clock ran fast (by up to 2x) inside a window; each window was rescaled to its true length, so times quoted here are approximate (roughly ± 30 s). Every gap of more than 30 s was re-transcribed on its own to confirm whether speech was missing. The opening minutes came back looped from the first five-minute window, so 0–4 min was re-transcribed in one-minute clips, and a left-over copy of that loop was removed during verification. Two short stretches (about 19:52–20:34 and 39:29–40:00) were spliced in from separate transcriptions; their duplicated sentences were removed. Equations are taken from the board frames, not from the transcript.

Total Internal Reflection and Its Applications

NCERT sections covered: 9.4, 9.4.1

Total internal reflection (NCERT 9.4)

An object under water sends rays up to the water-air surface. Going from denser (n_2) to rarer (n_1), each ray bends **away from the normal**, so $r > i$.

- As i is increased, r grows.
- At one particular angle of incidence, the refracted ray just grazes the surface: $r = 90^\circ$. This angle of incidence is the **critical angle** i_c (or C).
- For $i > i_c$, no light is refracted out. The ray is **reflected back into the same (denser) medium** and obeys the laws of reflection: angle of incidence = angle of reflection (e.g. 50° in, 50° out), with all rays in one plane.

Because the light is reflected entirely, and inside the medium it was already in, this is **total internal reflection (TIR)**. Unlike a mirror, it needs no silvered surface.

Conditions for TIR (board):

1. The ray must go from a **denser to a rarer** medium. Only then does it bend away from the normal, so r can reach 90° . Going rarer to denser, r can never be 90° .
2. The **angle of incidence must be greater than the critical angle** i_c .

Critical angle: the angle of incidence (in the denser medium) for which the angle of refraction is 90° .

Calculating the critical angle

Apply Snell's law at the critical angle, with each angle next to the index of its own medium:

$$n_2 \sin C = n_1 \sin 90^\circ = n_1 \Rightarrow \sin C = \frac{n_1}{n_2} = \frac{1}{n}, \quad C = \sin^{-1} \frac{1}{n}$$

where $n = n_2/n_1$.

- **Water:** $C = \sin^{-1}(1/1.33) \approx 49^\circ$.
- **Glass:** $C = \sin^{-1}(1/1.5) \approx 42^\circ$.

Applications of TIR (NCERT 9.4.1)

1. Sparkling of diamonds

- Diamond has a very high refractive index, so a very small critical angle (about 25° on the board).
- A ray entering a well-cut diamond meets face after face at more than 25° . It is **totally internally reflected many times** and comes out only through a few facets.
- A rough, uncut diamond hardly shines. Cutting many facets is what makes it sparkle.
- Glass ($C \approx 42^\circ$) traps light far less, which is the difference between a diamond and a glass imitation.

The board uses $n = 2.54$ for diamond. NCERT's Table 9.1 gives $n = 2.42$ and $i_c = 24.4^\circ$, so "about 25° " is right either way. The lecture also attributes the sparkle to the emerging beams arriving "in different phases". The standard explanation is the multiple TIR above, with the bright beams flashing as the stone or the eye moves.

2. Optical fibres

- A ray entering a glass fibre at a suitable angle meets the wall at more than the critical angle. It is **totally internally reflected again and again** and comes out at the far end, even when the fibre is bent.
- Optical fibres are used to **send signals**: the signal travels as light, very fast, with **negligible loss of energy**.
- The decorative lamp, with a bulb and a rotating coloured disc under a spray of glass fibres, works this way.
- *NCERT adds that each fibre has a core of higher refractive index surrounded by a cladding of lower index, and that fibres carry audio and video signals over long distances.*

3. Total reflecting prism

Board: an isosceles prism of crown glass ($n = 1.52$) with refracting angle 90° is called a total reflecting prism.

Light enters a face normally and meets the hypotenuse face at **45°**. Since $45^\circ > C \approx 42^\circ$, it is totally reflected. The prism is used to:

1. **deviate rays through 90°** (one TIR at the hypotenuse);
2. **deviate rays through 180°** (light enters the hypotenuse face, with two TIRs at the short faces); the image P, Q comes back at the same size;
3. **erect an inverted image without any deviation.**

(NCERT notes that for the 90° and 180° prisms, the critical angle of the prism material must be less than 45°, which is true for crown glass and dense flint glass.)

Natural examples of TIR

Mirage in deserts

- In extreme summer the air at the **ground is hot and has a lower refractive index**. The index increases with height.
- A ray from the top of a tree heading downwards passes from denser to rarer layers and bends away from the normal more and more. When $i > C$ it is **totally reflected** and curves up into the observer's eye.
- The observer sees an **inverted image of the tree** below the ground, as if reflected in a pool of water. This is also why a hot road looks wet in summer.
- **Air is dynamic**, so the image is not static: it **wavers**, like a reflection in rippling water.

Looming (not in the syllabus, as the teacher notes)

- In **extremely cold countries**, the air near the sea surface is cold and **denser**, with rarer air above.
- Rays from a ship going upwards pass from denser to rarer, bend away from the normal and are totally reflected back down to the observer.
- The ship appears **raised, floating in the air**. The lecture says the ship appears inverted; this kind of upper mirage can show it upright or inverted.

Note on this lecture's transcript: gemini-3.5-flash was in a 503 outage and then out of daily quota when this chapter was processed, so the audio was transcribed by gemini-3.5-flash-lite in overlapping five-minute windows, joined where the overlapping speech matches word for word. Lite's clock ran fast (by up to 2x) inside a window; each window was rescaled to its true length, so times quoted here are approximate (roughly ± 30 s). Every gap of more than 30 s was re-transcribed on its own to confirm whether speech was missing. The looming passage (about 24:16-24:57) was missing from the merged windows and was spliced in from a separate transcription of that stretch. Equations are taken from the board frames, not from the transcript.

Recap of Refraction and TIR; the World as Seen by a Fish

NCERT sections covered: 9.3, 9.4, 9.4.1

What this recording is

This 70-minute video is the teacher's unnumbered "Exp. for Lateral Displ., apparent depth, atmospheric refraction, TIR, Examples of TIR"; here it is numbered 14. It is **one continuous recording of the same lesson that lectures 10 and 13 were cut from**: the same words and the same board pages. Only its last seven minutes are new.

Time in this video	Topic	Full notes
00:00-06:00	Lateral displacement in a glass slab, $d = t \sin(i - r) / \cos r$	note 10
06:00-17:06	Real and apparent depth, the fish and fisherman, master formula $d' = d n_{\text{observer}} / n_{\text{medium}}$, stacked layers	note 10
17:06-27:37	Shift $t(1 - 1/n)$ due to a glass slab; the 3-mark board question	note 10
27:37-36:33	Atmospheric refraction: advance sunrise and delayed sunset, twinkling of stars	note 10
36:33-44:04	Total internal reflection, conditions, critical angle $\sin C = 1/n$	note 13
44:04-56:18	Applications: diamonds, optical fibres, total reflecting prisms	note 13
56:18-62:40	Mirage and looming	note 13
62:40-70:00	World as seen by a fish (below)	this note

The recording runs a few seconds ahead of lecture 10 at its start. It also includes the opening of the TIR lesson, which lecture 13's file begins just after, and a fuller description of looming. In cold regions the air in contact with the sea is cooler and denser, with rarer air above, so a ship appears **hanging in the air**, an optical illusion.

World as seen by a fish

Light from under the water (principle of reversibility).

- A bulb S at depth h under water sends rays up to the surface.
- Rays meeting the surface at less than the critical angle C escape.
- Rays beyond C are totally internally reflected back into the water.
- So **only a circle of the surface is lit**: its edge is where the rays meet the surface at exactly C .

From the right triangle with the bulb at depth h below the centre of the circle:

$$\tan C = \frac{r}{h} \Rightarrow r = h \tan C \quad (r \propto h)$$

What the fish sees. Run the rays the other way:

- Light from objects outside reaches the fish only through that circle.
- Even rays **grazing the air-water interface**, from objects near the surface at the horizon, enter the water and bend towards the normal.
- They reach the fish's eye at the critical angle C .

For water, $n = 1.33$ gives $C \approx 49^\circ$, so (board):

The rays grazing the air-water interface, starting from objects near the surface, enter water and reach the eye of the fish making critical angle $C = 49^\circ$ and become visible to the fish. To a fish the outside region is symmetrical. The whole outside world appears to it within a cone of apex angle $2C = 98^\circ$. The radius r on the surface is $r = h \tan C$, $r \propto h$.

So the Sun, trees, people on the bank, everything above the water, is squeezed into a cone of apex angle about 98° above the fish. Divers looking up see the same thing: **a bright circular window** of sunlight, with the rest of the surface looking dark or mirror-like.

Formula for numericals.

- $\sin C = \frac{1}{n}$, so $\cos^2 C = 1 - \sin^2 C = 1 - \frac{1}{n^2} = \frac{n^2 - 1}{n^2}$.
- Therefore $\tan C = \frac{\sin C}{\cos C} = \frac{1}{\sqrt{n^2 - 1}}$.

$$r = h \tan C = \frac{h}{\sqrt{n^2 - 1}}$$

(Check with water: $\sqrt{1.33^2 - 1} \approx 0.68$, so $r \approx 1.47 h$, which matches $\tan 49^\circ \approx 1.15$.)

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Magnifying Power of a Simple Microscope

NCERT sections covered: 9.7.1

Magnifying power of a simple microscope (NCERT 9.7.1)

A simple microscope is a single convex lens with the object **between O and F** (lecture 08). The image $A'B'$ is **virtual, erect and magnified**, on the same side as the object.

Definition. Let β be the angle the final image subtends at the eye. Let α be the angle the object subtends when it is placed directly in front of the eye at the **least distance of distinct vision**, D (25 cm).

$$m = \frac{\beta}{\alpha}$$

Both angles are small, so $\tan \alpha \approx \alpha$ and $\tan \beta \approx \beta$.

- With the eye at the lens, the image subtends the same angle as the object at distance u , so $\tan \beta = AB/u$.
- Placed at D , $\tan \alpha = AB/D$.
- u and D are both measured against the light, so both are negative and the signs cancel:

$$m = \frac{\tan \beta}{\tan \alpha} = \frac{AB/u}{AB/D} = \frac{D}{u}$$

In terms of f and v . From the lens formula $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$, put in the signs (u and v both negative, f positive):

$$\frac{1}{u} = \frac{1}{f} + \frac{1}{v} \quad \Rightarrow \quad m = D \left(\frac{1}{f} + \frac{1}{v} \right)$$

Here u and v are now magnitudes.

Case 1: final image at the least distance of distinct vision ($v = D$).

$$m = D \left(\frac{1}{f} + \frac{1}{D} \right) = 1 + \frac{D}{f}$$

Case 2: final image at infinity ($v = \infty$, normal adjustment).

$$m = D \left(\frac{1}{f} + \frac{1}{\infty} \right) = \frac{D}{f}$$

The magnifying power is **smaller in normal adjustment**, yet this mode is preferred. With the image at infinity the eye is **relaxed**, while focusing on an image at D strains it.

Numerical

Q. A convex lens of focal length 5 cm is used as a magnifying glass. Find its magnifying power if the final image is formed (i) at D , (ii) at infinity.

$$(i) \quad m = 1 + \frac{D}{f} = 1 + \frac{25}{5} = 6$$

$$(ii) \quad m = \frac{D}{f} = \frac{25}{5} = 5$$

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Magnifying Power of a Compound Microscope

NCERT sections covered: 9.7.1

Compound microscope: how the image is formed (NCERT 9.7.1)

A compound microscope has two convex lenses:

- a small **objective** of short focal length f_o , near the object;
- a larger **eyepiece** of focal length f_e , at the eye.

Memory hook: the objects viewed through a microscope are small, so the objective is the smaller lens.

Image formation.

1. The object AB is placed **between** f_o **and** $2f_o$ of the objective.
2. The objective forms a **real, inverted, magnified** image A'B' beyond $2f_o$.
3. A'B' must fall **between the eyepiece's focus F_e and its optical centre O_e** . The eyepiece then acts as a simple microscope on it and forms a large **virtual** final image A''B''.

Magnifying power

As for the simple microscope, $MP = \beta/\alpha$:

- β is the angle subtended at the eye by the final image;
- α is the angle the object would subtend at the least distance of distinct vision D .

The angles are small, and u_e is the distance of A'B' from the eyepiece:

$$MP = \frac{\tan \beta}{\tan \alpha} = \frac{-A'B'/(-u_e)}{AB/(-D)} = -\frac{A'B'}{AB} \cdot \frac{D}{u_e}$$

A'B' is inverted, so its height is negative. The board marks the two factors as m_o and m_e .

Magnification of the objective:

$$\frac{A'B'}{AB} = \frac{v_o}{u_o}$$

So:

$$\boxed{MP = -\frac{v_o}{u_o} \cdot \frac{D}{u_e}}$$

The eyepiece term. The lens formula for the eyepiece, with signs, gives $\frac{1}{u_e} = \frac{1}{f_e} + \frac{1}{v_e}$. Hence

$$MP = -\frac{v_o}{u_o} D \left(\frac{1}{f_e} + \frac{1}{v_e} \right)$$

Case 1: final image at D ($v_e = D$).

$$MP = -\frac{v_o}{u_o} \left(1 + \frac{D}{f_e} \right)$$

Case 2: final image at infinity ($v_e = \infty$, normal adjustment).

$$\text{MP} = -\frac{v_o}{u_o} \cdot \frac{D}{f_e}$$

In both cases $\text{MP} = m_o \times m_e$. The eyepiece is just a simple microscope (lecture 15) looking at the objective's already-magnified image.

Approximate formula

The object is placed very close to the objective's focus, so $u_o \approx f_o$. The first image forms very close to the eyepiece, so $v_o \approx L$, the **tube length** (the distance between the two lenses). Then

$$\text{MP} \approx -\frac{L}{f_o} \cdot \frac{D}{f_e} = -\frac{DL}{f_o f_e}$$

(In the recording the teacher reads this out as "-D f_o over f_e", a slip; the board has $-DL/(f_o f_e)$. NCERT defines L as the distance between the objective's second focal point and the eyepiece's first focal point. Taking $L \approx v_o$ is the lecture's simplification.)

For a large magnifying power, both f_o and f_e should be small, with $f_e > f_o$.

Q. You are given convex lenses of focal lengths 5 cm, 10 cm, 2 cm and 7 cm. Which two would you choose for a compound microscope, and which would be the objective?

Choose the two shortest, **2 cm and 5 cm**. Use $f_o = 2$ cm as the objective and $f_e = 5$ cm as the eyepiece.

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Magnifying Power of a Refracting Telescope

NCERT sections covered: 9.7.2

Refracting (astronomical) telescope (NCERT 9.7.2)

A telescope also uses two convex lenses, but the other way round from a microscope.

- The **objective is the bigger lens**, with a large focal length f_o and a large aperture.
Memory hook: the objects viewed through a telescope are big.
- The **eyepiece is the smaller lens**, with a short focal length f_e .

Image formation.

1. The object is very far away, so the rays from it arrive **parallel**.

2. The objective brings them to a real, inverted image A'B' in its **focal plane**, at distance f_o .
3. The eyepiece, at distance u_e from A'B', magnifies this into the final image A''B''.

Magnifying power

$$MP = \frac{\beta}{\alpha}$$

- β is the angle subtended at the eye by the final image.
- α is the angle subtended by the **object at the eye**. The object cannot be brought to the least distance of distinct vision. Because the tube is short compared with the object's distance, the angle at the objective equals the angle at the eye.

The angles are small, so:

$$\tan \beta = \frac{A'B'}{u_e}, \quad \tan \alpha = \frac{A'B'}{f_o}$$

$$MP = \frac{\tan \beta}{\tan \alpha} = -\frac{f_o}{u_e}$$

The minus sign comes from the sign convention (A'B' and u_e are negative).

The eyepiece term. The lens formula for the eyepiece, with signs, gives $\frac{1}{u_e} = \frac{1}{f_e} + \frac{1}{v_e}$. So

$$MP = -f_o \left(\frac{1}{f_e} + \frac{1}{v_e} \right)$$

Case 1: final image at the least distance of distinct vision ($v_e = D$).

$$MP = -f_o \left(\frac{1}{f_e} + \frac{1}{D} \right) = -\frac{f_o}{f_e} \left(1 + \frac{f_e}{D} \right)$$

Case 2: final image at infinity ($v_e = \infty$, normal adjustment).

$$MP = -\frac{f_o}{f_e} \quad (\text{magnitude } f_o/f_e, \text{ the NCERT result})$$

So $MP \propto f_o$ and $\propto 1/f_e$. Use an objective of **large** focal length and an eyepiece of **small** focal length.

Q. Given convex lenses of focal lengths 2 cm, 5 cm, 10 cm and 7 cm, which two make the best telescope?

$f_o = 10$ **cm** (largest) and $f_e = 2$ **cm** (smallest).

Length of the tube

The tube length is the distance between the two lenses:

$$L = f_o + u_e$$

In normal adjustment ($v_e = \infty$), the intermediate image must be at the eyepiece's focus, so $u_e = f_e$. The focal points of the two lenses coincide, and

$$\boxed{L = f_o + f_e}, \quad \text{MP} = \frac{f_o}{f_e}$$

Many exam numericals combine these two equations.

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Reflecting Telescope and the Cassegrain Telescope

NCERT sections covered: 9.7.2

Why a refracting telescope runs into trouble (NCERT 9.7.2)

The **aperture** of the objective (its opening, MN) should be **large**. A large aperture has a large **light-gathering power**, so distant, faint objects give a **bright** image. But a large lens brings problems.

1. Spherical aberration. "Aberration" means a defect or error.

- Parallel rays far from the axis, through the rim of the lens, are **marginal rays**. Rays close to the axis are **paraxial rays**.
- The two sets come to a focus at **different points**, so the image is **blurred**. A sharp image needs all rays to meet at one point.

2. Chromatic aberration. "Chroma" means colour.

- A lens disperses white light, so **violet is focused nearer the lens and red farther away**.
- Instead of a white image you get a **coloured band**.

3. Big lenses are difficult and costly to make. A large-aperture lens needs a lot of glass and is heavy.

Reflecting telescope

The objective is a **concave mirror of large aperture**, and this fixes these problems:

- A mirror only reflects, so there is no refraction, no dispersion and **no chromatic aberration**.
- A spherical mirror still has spherical aberration: its marginal and paraxial rays focus at different points. Using a **parabolic mirror** brings all parallel rays, paraxial or marginal, to **one point**, which **eliminates spherical aberration**.

The **Hubble Space Telescope** is a reflecting telescope.

Advantages of a reflecting telescope over a refracting telescope (board)

1. The image formed is **brighter**.
2. It is **free from chromatic aberration**.
3. **Spherical aberration is eliminated** by using a parabolic mirror.
4. Its **resolving power is greater** because of the large aperture, so finer detail can be seen.
5. It is **cheaper**.
6. It is **lighter in weight**, since a lens of the same aperture would need a great deal of glass.

Exam tip from the lecture: the main points are 1–4 (no chromatic aberration, no spherical aberration, a large aperture is possible). Lead with those, not cost and weight.

(NCERT adds that a mirror can be supported over its whole back surface, not just its rim, so mechanical support is much less of a problem.)

Cassegrain telescope (NCERT Fig. 9.26)

Construction (board):

- A **parabolic concave mirror** of large aperture and focal length is fitted at one end of a wide tube. This is the **objective mirror** (primary).
- A **convex mirror** is placed inside the tube as the **secondary mirror**.
- The rays reflected from the convex mirror pass through a **hole in the objective mirror** and reach the **eyepiece E**, which is a **convex lens**.

Working.

1. Parallel light from a distant object falls on the objective mirror and is reflected towards its focus.
2. Before reaching the focus, it strikes the secondary convex mirror.
3. The secondary reflects it back through the hole at the centre of the primary.
4. It forms the image, which is viewed through the eyepiece behind the primary.

Magnifying power:

$$m = \frac{f_o}{f_e}$$

where f_o is the focal length of the objective mirror and f_e that of the eyepiece.

The design is named after the physicist Cassegrain. Common exam questions ask you to draw the diagram, explain the working, and name the mirrors used (concave parabolic primary, convex secondary).

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